Unit: Unit 7: Factoring
Topic: Difference of Squares

Name: ________________
Date: ___________

Homework (Jalapeno)

Q1. Factor $x^2 - 16$
(A) $x^2 - 4^2$
(B) $x^2 - 4$
(C) $(x-4)(x+4)$
(D) $(x+4)(x-4)$
(E) $(x-4)^2$

Q2. In ancient Egypt, a rectangular field has a length of $x+5$ meters and a width of $x-5$ meters. Its area can be represented by which factored expression?
(A) $x^2 - 25$
(B) $(x+5)(x-5)$
(C) $x^2 + 25$
(D) $(x-5)^2$
(E) $(x+5)^2$

Q3. Factor $49 - y^2$
(A) $(7-y)(7+y)$
(B) $(7+y)(7-y)$
(C) $7^2 - y^2$
(D) $y^2 - 7^2$
(E) $(y+7)(y-7)$

Q4. The ancient Mayans used a formula $a^2 - b^2$ to calculate the area of their rectangular structures. What is the factored form of $a^2 - b^2$?
(A) $(a+b)(a-b)$
(B) $(a-b)(a+b)$
(C) $a^2 + b^2$
(D) $a^2 - 2ab$
(E) $b^2 - a^2$

Q5. What is the factored form of $x^2 - 9y^2$?
(A) $(x+3y)(x-3y)$
(B) $(x-3y)(x+3y)$
(C) $x^2 + 9y^2$
(D) $9y^2 - x^2$
(E) $3x^2 - y^2$

Q6. The ancient Greeks used the difference of squares to calculate the area of their public squares. If a square has an area of $x^2 - 4$, what is the factored form?
(A) $(x+2)(x-2)$
(B) $(x-2)(x+2)$
(C) $x^2 + 4$
(D) $x^2 - 2x$
(E) $x^2 + 2x$

Q7. Factor $25x^2 - 9$
(A) $(5x+3)(5x-3)$
(B) $(5x-3)(5x+3)$
(C) $25x^2 + 9$
(D) $9 - 25x^2$
(E) $5x^2 - 3$

Q8. The difference of squares formula was used by the ancient Babylonians to solve equations. What is the factored form of $x^2 - 1$?
(A) $(x+1)(x-1)$
(B) $(x-1)(x+1)$
(C) $x^2 + 1$
(D) $x^2 - x$
(E) $x^2 + x$

Open Ended Questions (FRQ)

Q9. Factor $x^2 - 36y^2$, and explain how the difference of squares formula applies to this expression.

Q10. A historical monument has a rectangular base with a length of $x+4$ meters and a width of $x-4$ meters. Factor the expression $x^2 - 16$ and use it to find the area of the base.

Chef's Tips

- Provide clear instructions for each question
- Encourage students to use historical context to understand the application of the difference of squares formula
- Allow students to use calculators or other aids for factoring